Summary Notes:
Introduction to Google Cloud Platform

Overview

Here is an overview of the topics covered in this session.

1. Cloud Functions
    a. Introduction to Cloud Functions
    b. Event-Driven Serverless Computing
    c. Use Cases and Examples
    d. Hands-on Activity: Creating and Deploying Cloud Functions
    e. Virtual Private Cloud (VPC)
2. Virtual Private Cloud (VPC)
    a. Overview of VPC and Subnets
    b. Configuring Firewalls and Routes
    c. Hands-on Activity: Setting Up a Custom VPC
3. Security and Identity Access Management (IAM)
    a. Overview of IAM
    b. Principles of GCP Security
    c. Managing Roles and Permissions
    d. Hands-on Activity: Configuring IAM Roles
4. Compute Engine
    a. Overview and Key Features
    b. Setting Up Virtual Machines
    c. Networking and Firewall Configuration
    d. VM Lifecycle Management
    e. Hands-on Activity: Launch and Connect to a VM
5. App Engine
    a. Introduction and Use Cases
    b. App Engine Architecture
    c. Standard vs Flexible Environments
    d. Deploying Applications
    e. Hands-on Activity: Deploying a Simple Web App

Session 1: Cloud Functions, VPC, Security, and IAM

1. Cloud Functions

Definition: Lightweight, event-driven, single-purpose functions that execute code without managing servers.

Benefits:

1. Reduces infrastructure costs with pay-as-you-go pricing.
2. Automatically scales based on demand.
3. Simplifies development and deployment processes.

Use Cases:

- Real-time Data Processing: Streaming IoT data.
- Webhooks: Automating responses to third-party events.
- Backend Operations: CRUD operations for web and mobile apps.
- File Processing: Resizing images, converting formats, and PDF generation.

2. Virtual Private Cloud (VPC)

Overview: Managed networking service for provisioning, connecting, and isolating resources.

Key Features:

- Subnets: Regional divisions within networks for resource organization.
- Firewall Rules: Controls traffic ingress and egress with stateful rules applied at the instance level.
- Routes: Maps IP ranges to destinations, allowing internal and external communication.

Hands-on Activity: Setting up a custom VPC with subnets and firewall rules.

3. Security and IAM

Definition: A system to define "Who can do what on which resource" in GCP.

Key Principles:

- Resource Hierarchy: Organization > Folders > Projects > Resources.
- Principle of Least Privilege: Grant minimum permissions necessary for tasks.
- Role-Based Access Control (RBAC): Assign roles at different levels (e.g., Basic, Predefined, Custom).

IAM Tools:

- Google Cloud Console
- gcloud Command-line Tool

Hands-on Activity: Configuring IAM roles and permissions.

Session 2: Compute Engine and App Engine

1. Compute Engine

Definition: Infrastructure as a Service (IaaS) providing virtual machines (VMs).

Features:

1. Flexible configurations for vCPU, memory, and storage.
2. Supports Linux and Windows OS.
3. Specialised VMs (e.g., preemptible, spot, and confidential VMs).

VM Setup:

1. **Configuration:** Choose regions, zones, and machine types.
2. **Disk Options:** Persistent disks for durability and local SSDs for high performance.
3. **Networking:** Configure subnets, IP allocation, and firewall rules.

VM Lifecycle:

1. **States:** Provisioning, running, repairing, and suspending.
2. **Key Actions:** Live migration, snapshots, and scaling.

2. App Engine

Definition: Serverless platform for building, deploying, and scaling web applications.

Architecture:

- Autoscaling adjusts instance count based on demand.
- Integrated with CI/CD pipelines for streamlined development.
- Built-in security features like Identity-Aware Proxy and TLS.

Deployment:

- Set up a project and enable the App Engine API.
- Prepare app code and add app.yaml.
- Deploy using gcloud app deploy and verify.